# Piper PA-46-310P Malibu (MALIBU)

Continental TSIO-520-BE, twin-turbocharged and intercooled, 310 hp · 6 seats · Typical PA-46-310P (1984–88, TSIO-520-BE, 4,100 lb, 5.5 psi cabin, hydraulic gear) POH procedures; analogue panel with King KFC 150 autopilot. Deice equipment and standby alternator where fitted; TIT and manifold pressure limits vary slightly with POH revision – use the placarded red lines.

## V-speeds & limits

| | |
|---|---|
| Vr | **70–74 KIAS (liftoff)** |
| Vx | **81 KIAS** |
| Vy | **110 KIAS** |
| Best glide | **90 KIAS (gear & flaps up, prop low RPM)** |
| Vfe | **165 KIAS (10°) / 130 KIAS (20°) / 116 KIAS (36° full)** |
| Vlo | **165 KIAS extend / 130 KIAS retract** |
| Vle | **165 KIAS** |
| Va | **135 KIAS (4,100 lb)** |
| Vno | **173 KIAS** |
| Vne | **203 KIAS** |
| Vs / Vso | **approx 69 / 58 KIAS (max weight)** |
| Vref | **80 KIAS final (full flaps); approx 75 KIAS short field** |
| Emergency descent | **165 KIAS gear down / 135 KIAS in turbulence** |

## Normal procedures

### Before Start

| | |
|---|---|
| Preflight inspection | **COMPLETE (fuel sumps, deice boots and static wicks, tyres, cowl and turbo area, no oil leaks)** |
| Weight & balance / takeoff data | **COMPUTED (4,100 lb max takeoff, 4,118 lb ramp)** |
| Passenger briefing | **COMPLETE (belts, airstair door, emergency exit, oxygen masks, no smoking)** |
| Seats, belts, shoulder harnesses | **ADJUST & LOCK** |
| Cabin door (upper & lower) / baggage doors | **CLOSED & LATCHED (DOOR AJAR annunciator out once power is on)** |
| Parking brake | **SET** |
| Landing gear selector | **DOWN** |
| Emergency gear extension lever | **NORMAL / STOWED (floor, between front seats)** |
| Throttle / propeller / mixture | **CLOSED / FULL FORWARD / IDLE CUTOFF** |
| Fuel selector | **FULLEST TANK (LEFT or RIGHT – OFF is detented, there is no BOTH)** |
| Fuel quantity | **CHECK both gauges (120 gal usable total)** |
| Avionics master | **OFF** |
| Battery master | **ON – voltage CHECK (approx 24 V)** |
| Annunciator panel | **PRESS TO TEST (all lights; gear lights three green)** |
| Cabin pressurisation dump switch | **NORMAL (not DUMP)** |
| Pressurisation controller | **SET field elevation + 500 ft; rate knob approx mid-range** |
| Oxygen | **CHECK pressure (approx 1,800 psi full) and quantity for flight; masks available** |
| Alternate static source | **OFF (normal)** |
| Beacon | **ON** |

## Engine Start

| | |
|---|---|
| Propeller area | **CLEAR** |
| Mixture | **FULL RICH** |
| Propeller | **FULL FORWARD (high RPM)** |
| Throttle | **FULL OPEN** |
| Fuel pump | **HI until fuel flow peaks (approx 3–5 sec), then OFF** |
| Throttle | **APPROX 1/4 OPEN** |

**▶ Engine hot or flooded?**

**No:**

**Yes:**

| | |
|---|---|
| Mixture | **IDLE CUTOFF** |
| Throttle | **FULL OPEN** |
| Mixture | **ADVANCE to FULL RICH as engine fires, throttle to idle** |
| Starter | **ENGAGE (release when engine fires; 30 sec max, then cool)** |
| Throttle | **1,000 RPM** |
| Oil pressure | **10 psi within 30 sec (normal 30–60 psi)** |
| Fuel pump | **OFF (LO if fuel flow fluctuates)** |
| Alternator switch | **ON – charging (ammeter / voltage approx 28 V)** |
| Standby alternator | **ON / ARMED (if fitted – STBY ALT light out)** |
| Avionics master | **ON** |
| Engine instruments | **CHECK (oil pressure, oil temp rising, fuel flow, TIT, CHT)** |

## Before Taxi

| | |
|---|---|
| Brakes | **CHECK on initial roll** |
| Flight instruments | **CHECK & SET (altimeter, HSI slaved, attitude erect, turn coordinator in turns)** |
| Avionics / GPS | **SET** |
| KFC 150 autopilot / electric trim | **PREFLIGHT TEST – disconnect checks, trim both directions – OFF** |
| Pitot heat / stall warning heat | **ON momentarily – CHECK (ammeter rise), then AS REQUIRED** |
| Propeller deice | **ON – ammeter in green, then AS REQUIRED (if fitted)** |
| Windshield hot plate | **ON – CHECK, then AS REQUIRED (if fitted)** |
| Surface deice (boots) | **SINGLE CYCLE – pneumatic pressure in green, boots inflate / deflate (if fitted)** |
| Elevator / rudder trim | **SET for takeoff** |
| Fuel selector | **CHECK fullest tank (LEFT or RIGHT)** |
| Flaps | **SET for takeoff (0° normal; 10° short field) – indicator CHECK** |

### Before Takeoff

| | |
|---|---|
| Parking brake | **SET** |
| Flight controls | **FREE & CORRECT** |
| Mixture | **FULL RICH** |
| Throttle | **1,700 RPM** |
| Magnetos | **CHECK L & R (150 RPM max drop, 50 RPM max difference)** |
| Propeller | **CYCLE 2–3 times (RPM drops, then returns)** |
| Engine instruments | **CHECK (oil pressure 30–60 psi, oil temp in green, fuel flow steady, TIT / CHT)** |
| Alternator / standby alternator | **CHECK charging, voltage approx 28 V** |
| Suction / pneumatic pressure | **CHECK in green (gyro suction approx 4.5–5.2 in Hg; deice pressure in green)** |
| Throttle | **IDLE check, then 1,000 RPM** |
| Fuel selector | **FULLEST TANK (feel for detent)** |
| Fuel pump | **LO** |
| Pressurisation | **Dump switch NORMAL; controller field elevation + 500 ft; door seal inflated, DOOR AJAR out** |
| Oxygen | **ON / AVAILABLE as required** |
| Propeller | **FULL FORWARD** |
| Flaps | **SET & INDICATING (0° or 10°)** |
| Trim | **SET takeoff range** |
| Landing gear selector | **DOWN – three green** |
| Ice protection / pitot heat | **AS REQUIRED** |
| Lights | **AS REQUIRED** |
| Transponder | **ALT** |
| Doors & windows | **LATCHED (DOOR AJAR out)** |
| Takeoff briefing | **COMPLETE (abort point; throttle slowly to red line – do not exceed max manifold pressure; gear up with positive rate and no usable runway, below 130 KIAS)** |

### After Takeoff / Climb

| | |
|---|---|
| Takeoff power | **FULL THROTTLE applied slowly – manifold pressure at or below red line (approx 36–38 in Hg per placard), 2,600 RPM** |
| Landing gear | **UP (positive rate, no usable runway, below 130 KIAS) – lights out, HYD PUMP light out** |
| Flaps | **UP (above approx 85 KIAS, obstacles cleared)** |
| Climb power | **REDUCE SLOWLY to approx 32 in Hg / 2,500 RPM (cruise climb)** |
| Mixture | **FULL RICH / climb fuel flow per POH schedule (TIT below 1,650°F)** |
| Fuel pump | **OFF above 1,000 ft AGL (LO if fuel flow fluctuates)** |
| Airspeed | **110 KIAS (Vy) / 120–130 KIAS cruise climb for cooling** |
| Pressurisation | **CHECK cabin climbing on rate, differential increasing (5.5 psi max); controller SET cruise altitude + 500 ft** |
| Engine instruments | **CHECK (CHT, TIT, oil temp 240°F max, oil pressure)** |
| Ice protection | **AS REQUIRED** |
| Landing / taxi lights | **OFF** |

### Cruise

| | |
|---|---|
| Cruise power | **SET (2,300–2,400 RPM, approx 26–31 in Hg; 75% max recommended) – change power slowly** |
| Mixture | **LEAN SLOWLY per POH cruise table – TIT 1,650°F max recommended cruise (1,750°F red line)** |
| Fuel selector | **ALTERNATE LEFT / RIGHT to keep tanks balanced – note time** |
| Fuel pump | **LO if fuel flow fluctuates (vapour suppression at altitude / hot fuel), otherwise OFF** |
| Fuel quantity / flow | **CHECK against plan** |
| Pressurisation | **CHECK cabin altitude, rate zero, differential 5.5 psi max (cabin approx 8,000 ft at FL250)** |
| Oxygen | **CHECK pressure; masks available (CABIN ALT warning approx 10,000 ft cabin)** |
| Engine instruments | **CHECK (oil 30–60 psi / 240°F max, CHT, TIT)** |
| Ice protection | **AS REQUIRED** |
| Altimeter / heading | **CHECK** |

### Descent

| | |
|---|---|
| Altimeter | **SET** |
| Pressurisation controller | **SET destination field elevation + 500 ft; rate approx 500 fpm (cabin ahead of aircraft)** |
| Power | **REDUCE SLOWLY (approx 1 in Hg per minute) – keep approx 18–20 in Hg minimum to hold cabin pressure** |
| Mixture | **ENRICH progressively (TIT / CHT monitored)** |
| Fuel selector | **FULLEST TANK (LEFT or RIGHT)** |
| Windshield heat / ice protection | **AS REQUIRED** |
| Airspeed | **BELOW 173 KIAS (Vno); plan below 165 KIAS before gear extension** |
| Approach briefing | **COMPLETE (Vref 80, flap setting, go-around power applied slowly)** |
| Seat belts / harnesses | **SECURE** |

### Before Landing

| | |
|---|---|
| Pressurisation | **CHECK cabin altitude at or above field elevation, differential approaching zero** |
| Fuel selector | **FULLEST TANK (LEFT or RIGHT)** |
| Fuel pump | **LO** |
| Mixture | **FULL RICH (as required at high field elevation)** |
| Airspeed | **BELOW 165 KIAS** |
| Landing gear | **DOWN – three green, HYD PUMP light out** |
| Flaps | **10° below 165 KIAS; 20° below 130 KIAS; 36° on final below 116 KIAS** |
| Propeller | **FULL FORWARD (on final)** |
| Autopilot / yaw damper | **OFF** |
| Airspeed | **80 KIAS final (full flaps; approx 75 KIAS short field; add 10–15 KIAS with residual ice)** |
| Landing light | **ON** |
| Landing gear | **THREE GREEN – CONFIRMED on final** |

### After Landing

| | |
|---|---|
| Flaps | **UP** |
| Pressurisation | **CONFIRM depressurised (differential zero; dump if required)** |
| Fuel pump | **OFF** |
| Ice protection | **OFF (pitot heat, prop deice, windshield heat, boots)** |
| Landing light / strobes | **AS REQUIRED** |
| Transponder | **STBY / as required** |
| Trim | **SET takeoff** |

### Shutdown

| | |
|---|---|
| Parking brake | **SET** |
| Throttle | **1,000 RPM – turbocharger cool-down 3–5 min (TIT stabilised / falling)** |
| Avionics master | **OFF** |
| Electrical equipment / lights | **OFF** |
| Oxygen | **OFF** |
| Mixture | **IDLE CUTOFF** |
| Magnetos | **OFF (after propeller stops)** |
| Alternator / standby alternator / battery master | **OFF** |
| Control lock | **INSTALL** |
| Chocks / tiedowns / covers | **AS REQUIRED** |

## Abnormal procedures

### Alternator Failure / Standby Alternator

ALT INOP annunciator, ammeter zero or voltage below approx 25 V

Alternator switch ..... **OFF, then ON (reset)**

Alternator field circuit breaker ..... **CHECK / RESET once**

**▶ Alternator restored?**

**Yes:**

- Ammeter / voltmeter ..... **MONITOR**

**No – standby alternator fitted:**

- Alternator switch ..... **OFF**
- Standby alternator ..... **ON – STBY ALT ON light; voltage CHECK**
- Electrical load ..... **REDUCE to within standby alternator capacity (approx 20 A: essential avionics, pitot heat; windshield heat and prop deice OFF)**
- Landing gear / flaps ..... **PLAN early gear extension – hydraulic pump and flaps draw heavily**
- Deferred: Main alternator failed – standby alternator, limited electrical load; land as soon as practical. ..... **NOTED**

**No – battery only:**

- Alternator switch ..... **OFF**
- Non-essential electrical load ..... **OFF (avionics to minimum, deice OFF, lights OFF)**
- Landing gear ..... **EXTEND EARLY while battery power remains (free-fall extension if the pump fails)**
- Pressurisation ..... **UNAFFECTED (pneumatic), but CABIN ALT warning and gear lights may fail – monitor**
- Deferred: Alternator failed – battery only; extend gear early, flaps-up landing possible, land as soon as practical. ..... **NOTED**

### Low Oil Pressure

Oil pressure below 30 psi in flight

Oil pressure / oil temperature ..... **CHECK and cross-check**

**▶ Oil temperature rising or pressure below 10 psi?**

**Yes – genuine:**

- Power ..... **REDUCE to minimum required (turbochargers are oil-lubricated – expect failure)**
- Airport ..... **DIVERT to nearest suitable, stay within glide of a landing area**
- Altitude ..... **HOLD altitude as long as practicable for glide range**
- Deferred: Low oil pressure – land as soon as possible, expect possible engine failure. ..... **NOTED**

**No – gauge suspect:**

- Flight ..... **CONTINUE, monitor oil temperature and pressure**
- Deferred: Oil pressure indication suspect – maintenance check required. ..... **NOTED**

### Turbocharger Failure / Manifold Pressure Loss

Manifold pressure falls or cannot be obtained, fuel flow high for the MP, rough running, possible exhaust smell

Throttle ..... **REDUCE, then re-set smoothly**

Mixture ..... **LEAN to smooth running at the manifold pressure available (fuel flow too high for MP)**

Engine instruments ..... **CHECK (TIT, CHT, oil pressure and temperature)**

Exhaust leak indications ..... **CHECK (smell, smoke, discoloured cowl, cabin CO) – exhaust gas in cowling is a fire hazard**

**▶ Exhaust leak or fire suspected?**

**Yes:**

- Power ..... **MINIMUM required**
- Cabin heat ..... **OFF (CO risk)**
- Landing ..... **NEAREST SUITABLE AIRPORT – be ready for Engine Fire in Flight**
- Deferred: Turbocharger / exhaust failure – fire hazard, land as soon as possible, no go-around. ..... **NOTED**

**No – power loss only:**

- Altitude ..... **DESCEND to where naturally-aspirated power is adequate (approx 25 in Hg available near sea level, less with altitude)**
- Pressurisation ..... **EXPECT cabin to climb (turbo bleed air lost) – oxygen ON above 10,000 ft cabin, descend**
- Deferred: Turbocharger failed – reduced power, no pressurisation, longer takeoff distance; land as soon as practical. ..... **NOTED**

## High TIT / Overboost

TIT above 1,750°F red line or manifold pressure above red line

**▶ Indication?**

**Overboost (MP above red line):**

- Throttle — **REDUCE below red line, re-apply slowly**
- Oil temperature — **CHECK (cold oil slows the controller – warm up before full power)**

Momentary overboost of approx 1 in Hg for a few seconds is tolerated; sustained or large overboost requires a maintenance inspection before further flight.

Deferred: Overboost occurred – record peak MP and duration, maintenance inspection required. — **NOTED**

**High TIT:**

- Mixture — **ENRICH until TIT below 1,650°F**
- Throttle — **REDUCE if TIT remains high**
- Fuel flow — **CHECK against POH for the power set (low flow – fuel pump LO)**

**▶ TIT within limits?**

**Yes:**

- Flight — **CONTINUE, richer mixture**

**No:**

- Power — **REDUCE to the highest setting that keeps TIT below 1,650°F**

Deferred: High TIT – reduced power, suspect fuel system or TIT probe; maintenance check. — **NOTED**

## Fuel Flow Fluctuation / Low Fuel Pressure

- Fuel pump — **LO (vapour suppression); HI only if fuel flow is very low – may flood engine at low power**
- Fuel selector — **SWITCH TANKS (check quantity in the tank being used)**
- Mixture — **ADJUST for smooth running**
- Engine instruments — **CHECK (fuel flow, TIT, CHT)**

**▶ Fuel flow steady and engine smooth?**

**Yes:**

Deferred: Fuel flow fluctuation cleared – leave fuel pump LO; report to maintenance. — **NOTED**

**No:**

- Power — **REDUCE to smoothest setting; lower altitude helps vapour**

Deferred: Fuel flow unstable – land as soon as practical at nearest suitable airport. — **NOTED**

## Pressurisation Controller Fault / Cabin Not Holding

Cabin altitude or rate not following the controller; differential abnormal

- Cabin pressurisation dump switch — **NORMAL**
- Pressurisation controller — **RESELECT cabin altitude / rate; cycle rate knob**
- Manifold pressure — **ABOVE approx 20 in Hg (bleed air inflow needs power)**
- Door seal / DOOR AJAR — **CHECK (hiss at door, light out)**
- Cabin differential — **CHECK – 5.5 psi max (safety valve relieves above)**

**▶ Cabin response?**

**Cabin following again:**

- Pressurisation — **MONITOR closely**

Deferred: Pressurisation controller intermittent – maintenance check. — **NOTED**

**Cabin climbing:**

- Oxygen — **ON if cabin above 10,000 ft**
- Altitude — **DESCEND to keep cabin below 10,000 ft; Rapid Decompression procedure if CABIN ALT warning**

Deferred: Pressurisation failed – unpressurised or limited altitude, oxygen as required. — **NOTED**

**Differential too high:**

- Altitude / power — **DESCEND; reduce power to reduce inflow**
- Before landing — **DUMP below 10,000 ft so the cabin is depressurised at touchdown**

Deferred: Outflow valve fault – dump before landing; do not open the door until differential is zero. — **NOTED**

## Door Seal Failure / Door Open in Flight

DOOR AJAR light, loud hiss at door, cabin will not hold pressure

Aircraft ..... **CONTROLLABLE – fly the airplane**

Airspeed ..... **REDUCE to approx 110–120 KIAS**

Belts / loose items ..... **SECURE; passengers away from the door**

**▶ Door seal only (door latched) or door unlatched?**

**Seal failed, door latched:**

Altitude ..... **DESCEND – cabin cannot hold pressure; oxygen if cabin above 10,000 ft**

Deferred: Door seal failed – unpressurised, remain at or below 10,000–12,500 ft, maintenance required. ..... **NOTED**

**Door unlatched:**

Cabin pressurisation ..... **DUMP (do not pressurise)**

Do not attempt to close or re-latch the door in flight. Expect noise and buffet.

Deferred: Cabin door unlatched – unpressurised, return and land as soon as practical, normal approach speed. ..... **NOTED**

## Landing Gear Unsafe / Fails to Retract or Extend

Red gear warning light, fewer than three green, or HYD PUMP light stays on

Airspeed ..... **BELOW 165 KIAS (below 130 KIAS to retract)**

Gear indicator bulbs ..... **CHECK (press to test / dim switch) – navigation lights dim the gear lights**

Hydraulic pump circuit breaker ..... **CHECK / RESET once**

Landing gear selector ..... **CYCLE once to the desired position**

**▶ Gear in desired position?**

**Yes:**

Gear indication ..... **CONFIRM three green / lights out**

Deferred: Gear system fault – do not retract after landing; hydraulic / gear inspection required. ..... **NOTED**

**No – failed to retract:**

Landing gear selector ..... **DOWN – leave down**

Hydraulic pump ..... **If HYD PUMP light stays on, PULL hydraulic pump circuit breaker (pump running against a leak)**

Airspeed ..... **BELOW 165 KIAS**

Deferred: Gear will not retract – gear down, 165 KIAS max, reduced range; land as soon as practical. ..... **NOTED**

**No – failed to extend:**

Use Emergency Gear Extension (Free Fall).

## Flap Failure / Asymmetry

Flap selector ..... **STOP at current position – do not continue extension**

Flap circuit breaker ..... **CHECK / RESET once**

**▶ Flaps operable and symmetric?**

**Yes:**

Flap setting ..... **AS REQUIRED**

**No:**

Landing ..... **AT CURRENT FLAP SETTING – approach approx 90 KIAS flaps up, longer runway**

Deferred: Flaps inoperative / asymmetric – flaps-up or partial-flap landing, approach approx 90 KIAS, increased landing distance. ..... **NOTED**

## Propeller Overspeed

RPM above 2,600 and not responding to the prop lever

Throttle ..... **REDUCE**

Propeller ..... **FULL AFT (low RPM)**

Airspeed ..... **REDUCE (pitch up) to bring RPM within limits**

Oil pressure ..... **CHECK (governor uses engine oil)**

Power ..... **SET minimum that keeps RPM at or below 2,600**

Deferred: Propeller governor fault – control RPM with throttle and airspeed, land as soon as practical. ..... **NOTED**

### Vacuum / Pneumatic Pressure Loss

Suction gauge low, VAC annunciator, or deice pressure low

Suction / pneumatic pressure gauges ..... **CHECK (dual pumps – one failed leaves reduced capacity)**

Attitude / heading indicators ..... **CROSS-CHECK with turn coordinator, compass, GPS, electric standby instruments**

▶ **Suction and deice pressure?**

**One pump failed – suction still in green:**

- Surface deice ..... **USE SPARINGLY – single pump; exit icing conditions**
- Deferred: One air pump failed – gyros on remaining pump, limited boot capacity; avoid icing. ..... **NOTED**

**Suction lost:**

- Failed gyro instruments ..... **COVER**
- KFC 150 autopilot / flight director ..... **OFF (air-driven attitude gyro feeds the autopilot – expect disconnect)**
- Surface deice ..... **INOPERATIVE – exit icing immediately**
- IMC ..... **EXIT / AVOID – partial panel**
- Deferred: Vacuum / pneumatic failure – partial panel, no autopilot, boots inoperative; prefer VMC approach. ..... **NOTED**

### Pitot / Static Source Blockage

Airspeed erratic or zero, altimeter / VSI frozen

Pitot heat ..... **ON**

Attitude and power ..... **FLY KNOWN pitch / power settings; GPS ground speed as cross-check**

▶ **Static system blocked?**

**Yes:**

- Cabin pressurisation ..... **DUMP / depressurise (below 10,000 ft) before using cabin static**
- Alternate static source ..... **ON (cabin) – only with cabin unpressurised; expect small altimeter / airspeed errors**
- Deferred: Static source blocked – alternate static in use, cabin depressurised, altimeter error on approach. ..... **NOTED**

**No – pitot only:**

- Airspeed indicator ..... **DISREGARD until heat clears the blockage; use attitude, power and GPS**
- Deferred: Airspeed unreliable – approach by attitude and power, add margin, longer runway. ..... **NOTED**

## Emergency procedures

### Engine Failure During Takeoff Roll

Throttle (memory) ..... **CLOSED**

Brakes (memory) ..... **AS REQUIRED – stop straight ahead**

Mixture ..... **IDLE CUTOFF**

Fuel selector ..... **OFF**

Magnetos ..... **OFF**

Battery master / alternator ..... **OFF**

### Engine Failure After Takeoff

Airborne at low altitude

**LAND IMMEDIATELY**

Airspeed (memory) ..... **90 KIAS (best glide, gear & flaps up); 80–85 KIAS with gear / flaps down**

Fuel pump (memory) ..... **HI (if fuel flow low); fuel selector SWITCH TANKS if time permits**

▶ **Runway remaining to land on?**

**Yes:**

- Landing gear ..... **DOWN (if retracted)**
- Flaps ..... **AS REQUIRED**
- Land ..... **STRAIGHT AHEAD**

**No – land ahead off-field:**

- Landing gear ..... **DOWN on firm surface / UP on rough terrain or water**
- Flaps ..... **DOWN (36°) when landing assured**
- Cabin pressurisation ..... **DUMP (so the door opens after landing)**
- Fuel selector ..... **OFF**
- Mixture ..... **IDLE CUTOFF**
- Magnetos ..... **OFF**
- Battery master ..... **OFF (after gear position set)**

Land straight ahead with only small changes of direction. Do not attempt a turn back.

## Engine Failure in Flight

Loss of power with altitude available

**LAND AS SOON AS POSSIBLE**

| | |
|---|---|
| Airspeed (memory) | **90 KIAS (best glide, gear & flaps up)** |
| Fuel selector (memory) | **SWITCH TANKS (feel for detent)** |
| Fuel pump (memory) | **LO; HI if fuel flow reads zero (lean if engine floods)** |
| Mixture (memory) | **FULL RICH, then lean as required** |
| Propeller (memory) | **FULL AFT (low RPM) to extend glide** |
| Oxygen | **ON if cabin altitude climbs above 10,000 ft (pressurisation is lost without engine power)** |

**▶ Restart attempt practicable?**

**Yes – restart:**

| | |
|---|---|
| Magnetos | **BOTH – check** |
| Alternate induction air | **OPEN (if fitted / induction icing suspected)** |
| Throttle | **1/4 OPEN** |
| Starter | **ENGAGE if propeller has stopped** |

**▶ Engine restarted?**

**Yes:**

| | |
|---|---|
| Power | **AS REQUIRED – apply slowly** |
| Fuel pump | **LO if required** |
| Deferred: Engine restarted after in-flight failure – land as soon as possible, suspect fuel or induction system. | **NOTED** |

**No:**

Continue with forced landing.

**No – forced landing:**

Forced landing without power:

| | |
|---|---|
| ATC / ELT | **MAYDAY, squawk 7700** |
| Seats / belts / harnesses | **SECURE; passengers briefed** |
| Landing gear | **DOWN on firm surface (below 165 KIAS; use emergency extension if hydraulic pump fails) / UP on rough terrain or water** |
| Flaps | **DOWN when landing assured** |
| Airspeed | **80–85 KIAS on final** |
| Cabin pressurisation | **DUMP** |
| Fuel selector | **OFF** |
| Mixture | **IDLE CUTOFF** |
| Magnetos | **OFF** |
| Battery master / alternator | **OFF (after gear and flaps set)** |
| Cabin door | **UNLATCH before touchdown** |

## Engine Fire in Flight

**LAND IMMEDIATELY**

| | |
|---|---|
| Fuel selector (memory) | **OFF** |
| Mixture (memory) | **IDLE CUTOFF** |
| Fuel pump (memory) | **OFF** |
| Throttle (memory) | **CLOSED** |
| Cabin pressurisation (memory) | **DUMP (bleed air OFF to cabin); cabin heat / defroster OFF** |
| Oxygen | **ON if smoke in cabin / cabin altitude above 10,000 ft** |
| Airspeed | **INCREASE (up to 165 KIAS) to help extinguish fire** |
| Landing gear | **DOWN while battery power remains (emergency extension if needed)** |
| Forced landing | **EXECUTE (do not attempt restart)** |

## Engine Fire During Start

Starter (memory) ..... CONTINUE CRANKING

**▶ Engine starts?**

Yes:

- Throttle ..... 1,700 RPM for a few minutes
- Engine ..... SHUT DOWN and inspect for damage

No:

- Mixture ..... IDLE CUTOFF
- Throttle ..... FULL OPEN
- Fuel pump ..... OFF
- Fuel selector ..... OFF
- Starter ..... CONTINUE CRANKING to draw fire into engine
- Magnetos ..... OFF
- Battery master / alternator ..... OFF
- Fire extinguisher ..... OBTAIN / USE
- Aircraft ..... EVACUATE

## Electrical / Cabin Fire and Smoke

**LAND AS SOON AS POSSIBLE**

Battery master / alternator / standby alternator (memory) ..... OFF

Avionics master (memory) ..... OFF

Cabin heat / defroster / vents ..... OFF / CLOSED

Fire extinguisher ..... ACTIVATE (then ventilate cabin)

Oxygen ..... CREW MASKS ON if smoke persists (keep oxygen away from flames)

**▶ Fire out and electrical power required?**

Yes:

- Circuit breakers ..... CHECK – do not reset faulted
- Battery master ..... ON
- Alternator / avionics / equipment ..... ON one at a time, check for smoke
- Deferred: Electrical fire – suspect circuits isolated, affected systems unavailable. ..... NOTED

No:

Landing gear hydraulic pump and flaps are electric: plan an emergency free-fall gear extension and a flaps-up landing.

- Deferred: Electrical power off – emergency gear extension, flaps-up landing, no landing light, no gear lights. ..... NOTED

Smoke evacuation:

Descent ..... TO 10,000 ft or below if practicable

Cabin pressurisation ..... DUMP (below 10,000 ft) to ventilate

Fresh air vents / storm window ..... OPEN

## Emergency Descent

Throttle (memory) ..... CLOSED (reduce smoothly if time permits)

Propeller (memory) ..... FULL FORWARD

Landing gear (memory) ..... DOWN (below 165 KIAS)

Flaps (memory) ..... 10° (below 165 KIAS)

Airspeed (memory) ..... 165 KIAS (smooth air) / 135 KIAS (turbulence)

Oxygen ..... ON if cabin altitude above 10,000 ft

Mixture ..... ENRICH as required

ATC ..... ADVISE – emergency descent

Level-off ..... 10,000 ft or MEA – restore power slowly; gear up below 130 KIAS if continuing

## Rapid Decompression / Cabin Altitude High

CABIN ALT warning (approx 10,000 ft cabin), ears / fog / loud decompression, or cabin climbing uncontrolled

**LAND AS SOON AS PRACTICAL**

| | |
|---|---|
| Oxygen masks (crew) (memory) | **DON – oxygen ON, 100%** |
| Passenger oxygen (memory) | **ON – masks DON** |
| Emergency descent (memory) | **INITIATE to 12,500 ft or below (10,000 ft preferred)** |
| ATC | **ADVISE** |
| Pressurisation | **CHECK: dump switch NORMAL, controller set below flight altitude, manifold pressure above approx 20 in Hg, DOOR AJAR out** |

**▶ Cabin pressurisation recovered?**

**Yes:**

| | |
|---|---|
| Cabin altitude | **MONITOR; stay at an altitude the cabin holds below 10,000 ft** |
| Deferred: Pressurisation fault – low cruise altitude, oxygen used, maintenance inspection required. | **NOTED** |

**No:**

| | |
|---|---|
| Altitude | **REMAIN at or below 12,500 ft (10,000 ft without oxygen for passengers)** |
| Deferred: Unpressurised – remain low, oxygen quantity limited, no high-altitude diversion. | **NOTED** |

## Emergency Gear Extension (Free Fall)

Gear does not extend hydraulically / HYD PUMP light stays on

| | |
|---|---|
| Airspeed | **REDUCE to approx 100 KIAS (below 165 KIAS Vle; slower aids free fall)** |
| Hydraulic pump circuit breaker | **PULL** |
| Landing gear selector | **DOWN** |
| Emergency gear extension lever | **PULL / EMERGENCY position (floor, between front seats) – hydraulic pressure released, gear free-falls** |
| Aircraft | **YAW left and right to assist gear into locks** |
| Landing gear indicator | **THREE GREEN** |

Do not reset the hydraulic pump circuit breaker or move the gear selector up after an emergency extension. Leave the lever in the emergency position and the gear down.

**▶ Three green obtained?**

**Yes:**

| | |
|---|---|
| Deferred: Gear extended by free fall – do not retract; gear and hydraulic system inspection required. | **NOTED** |

**No:**

Continue to Gear-Up / Partial Gear Landing.

## Gear-Up / Partial Gear Landing

Gear cannot be confirmed down

| | |
|---|---|
| Runway | **LONGEST HARD SURFACE available, emergency services advised** |
| Passenger briefing | **COMPLETE (belts tight, brace, exits)** |
| Landing gear | **UP (or as it is – do not cycle)** |
| Flaps | **DOWN (36°)** |
| Approach | **NORMAL, 80 KIAS** |
| Fuel pump | **OFF** |
| Cabin pressurisation | **DUMP** |
| Cabin door | **UNLATCHED** |

On short final:

| | |
|---|---|
| Mixture | **IDLE CUTOFF** |
| Fuel selector | **OFF** |
| Magnetos | **OFF** |
| Battery master / alternator | **OFF** |
| Touchdown | **MINIMUM SPEED, slightly nose high, wings level** |
| Aircraft | **EVACUATE once stopped** |

## Inadvertent Icing Beyond Capability

Ice accumulating faster than the deice system clears it, or aircraft not approved for icing

**LAND AS SOON AS PRACTICAL**

| | |
|---|---|
| Pitot heat / stall warning heat (memory) | **ON** |
| Propeller deice / windshield heat (memory) | **ON (if fitted)** |
| Surface deice (boots) (memory) | **CYCLE at first sign of ice – repeat as required (if fitted)** |
| Exit icing (memory) | **TURN BACK / CLIMB / DESCEND – ATC advised, request new altitude** |
| Power | **INCREASE to hold airspeed; manifold pressure monitored (induction ice – alternate air if fitted)** |
| Autopilot | **OFF – hand fly, note control feel and trim** |
| Airspeed | **MAINTAIN higher than normal; no abrupt pitch changes** |
| Cabin heat / defroster | **FULL ON – windshield hot plate for forward view** |

**▶ Residual ice on airframe for landing?**

**No:**

| | |
|---|---|
| Ice protection | **OFF when clear** |

**Yes:**

| | |
|---|---|
| Flaps | **10° maximum for landing** |
| Approach speed | **INCREASE 10–15 KIAS (approx 90–95 KIAS); expect stall warning unreliable** |
| Landing | **LONGER RUNWAY, minimal flare, no steep turns** |
| Deferred: Residual airframe ice – flaps 10°, approach 90–95 KIAS, longer landing distance. | **NOTED** |

## Spin Recovery

Intentional spins prohibited – inadvertent spin entry

| | |
|---|---|
| Throttle (memory) | **IDLE** |
| Ailerons (memory) | **NEUTRAL** |
| Rudder (memory) | **FULL OPPOSITE to rotation** |
| Control wheel (memory) | **FULL FORWARD (release back pressure)** |
| Rudder (memory) | **NEUTRAL as rotation stops** |
| Recovery | **SMOOTH pull-out from dive; do not exceed 203 KIAS (Vne) or load limits** |
| Flaps / gear | **UP during recovery if extended** |
| Power | **RESTORE slowly once in level flight** |

Prototype content. Verify against the operator's approved checklist and the POH/AFM before any use. AeroE Checklist version 1.8 (build c9e52546) · 2026-11-07